

- 6 (a) (i) A packet (or quantum) of electromagnetic radiation energy; the quantum is given by $h \times \text{frequency of radiation}$ or the quantum is proportional to the frequency of radiation. 1
1

(ii)
$$\text{Power} = \frac{\text{Energy of UV photons incident on the surface}}{\text{time}}$$

Intensity $I \times \text{surface area } A = \frac{\text{no. of photons } N \times \text{energy of a photon } (hf)}{\text{time } t}$ 1

i.e. $I \times A = \frac{N(hf)}{t} = \frac{N\left(\frac{hc}{\lambda}\right)}{t}$, where λ is wavelength of the UV radiation. 1

rate of incidence of photons, $\frac{N}{t} = \frac{I \times A}{\left(\frac{hc}{\lambda}\right)} = \frac{210 \times 12 \times 10^{-6}}{\left(\frac{6.63 \times 10^{-34} (3.00 \times 10^8)}{254 \times 10^{-9}}\right)}$

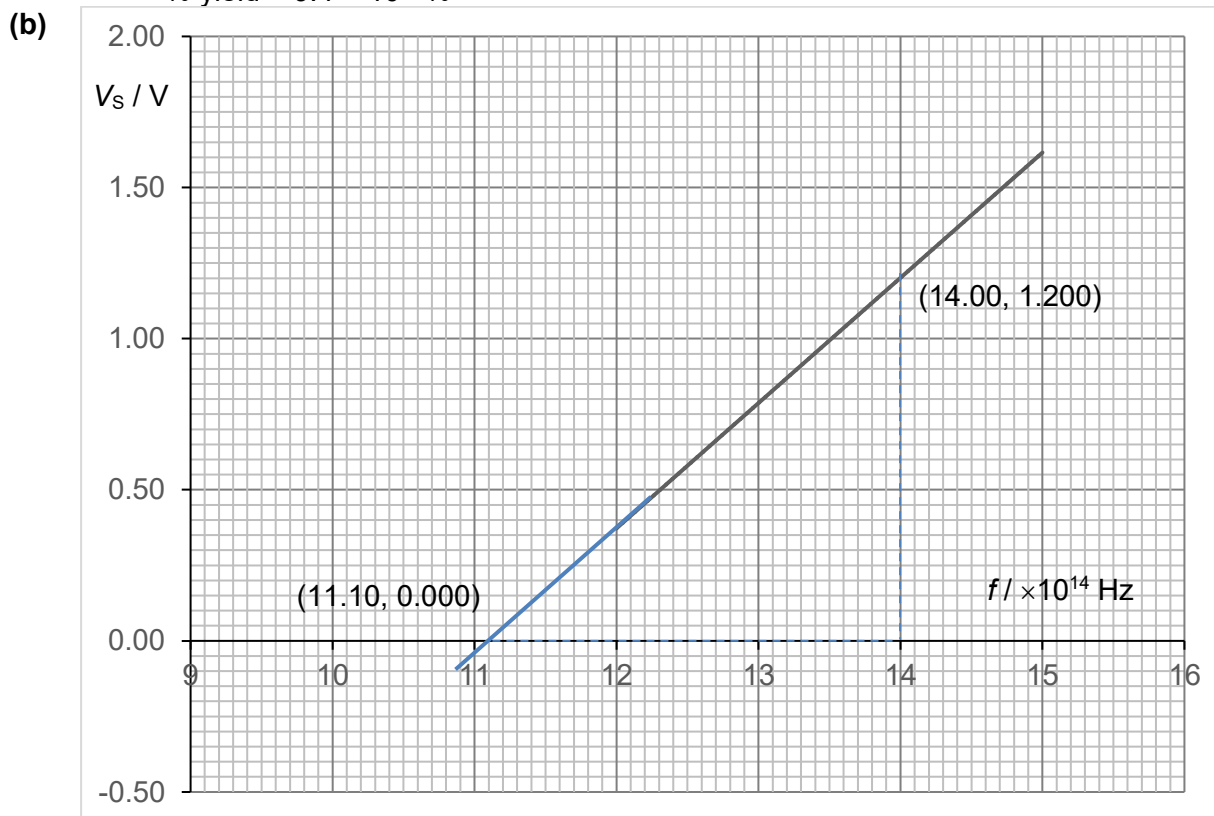
$$\frac{N}{t} = 3.218 \times 10^{15} = 3.2 \times 10^{15}$$

(iii) Current $I_c = \frac{Q}{t} = \frac{n_e e}{t}$ 1

Rate of emission of electrons, $\frac{n_e}{t} = \frac{I_c}{e} = \frac{4.80 \times 10^{-10}}{1.60 \times 10^{-19}} = 3.00 \times 10^9$

$$\frac{3.00 \times 10^9}{3.2 \times 10^{15}} = 9.4 \times 10^{-7}$$
 1

% yield = $9.4 \times 10^{-5} \%$



(b) (i) Photoelectric equation:

$$eV_s = hf - \Phi$$

$$V_s = \frac{h}{e}f - \frac{\Phi}{e} \quad 1$$

$$\text{Gradient of } V_s - f \text{ graph, } \frac{h}{e} = \frac{1.200 - 0.000}{(14.00 - 11.10) \times 10^{14}} = 4.138 \times 10^{-15} \quad 1$$

$$h = 4.138 \times 10^{-15} \times 1.60 \times 10^{-19} = 6.62 \times 10^{-34} \text{ J s} \quad 1$$

(accepted range: 6.6-6.7?)

(ii) Extrapolating the graph, threshold frequency = $f_0 = 11.10 \times 10^{14} \text{ Hz}$ 1

$$\Phi = hf_0 = (6.62 \times 10^{-34})(11.10 \times 10^{14}) / 1.60 \times 10^{-19} = 4.59 \text{ eV} \quad 1$$

OR:

$$V_s = \frac{h}{e}f - \frac{\Phi}{e}$$

$$\frac{\Phi}{e} = \frac{h}{e}f - V_s$$

Sub gradient, $(14.00 \times 10^{14}, 1.200)$:

$$\frac{\Phi}{e} = (4.138 \times 10^{-15})(14.00 \times 10^{14}) - 1.200 = 4.59 \text{ V}$$

$$\Phi = 4.59 \text{ eV}$$

[Total: 11]

2021

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

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(a)

(i)

CM